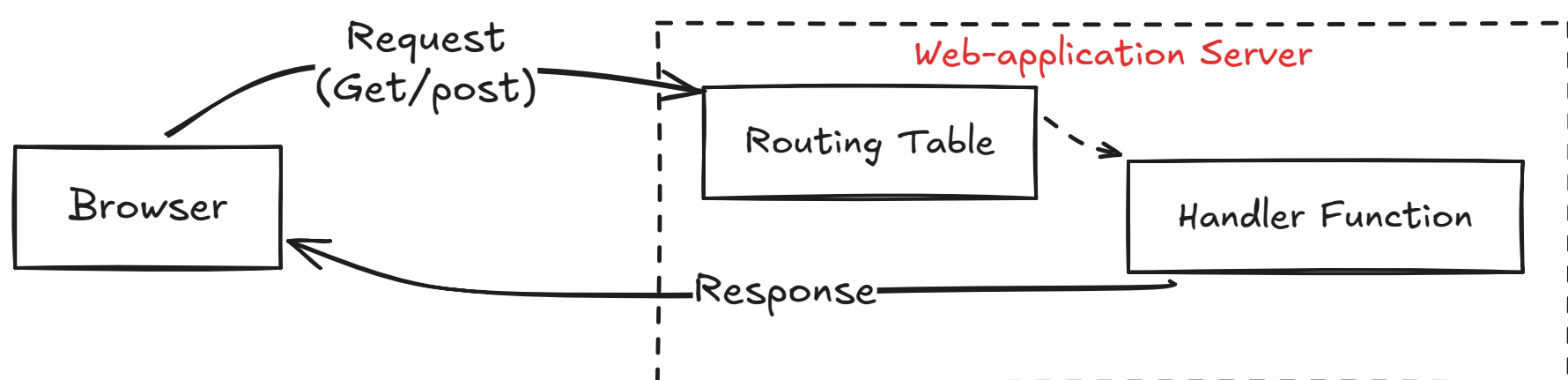
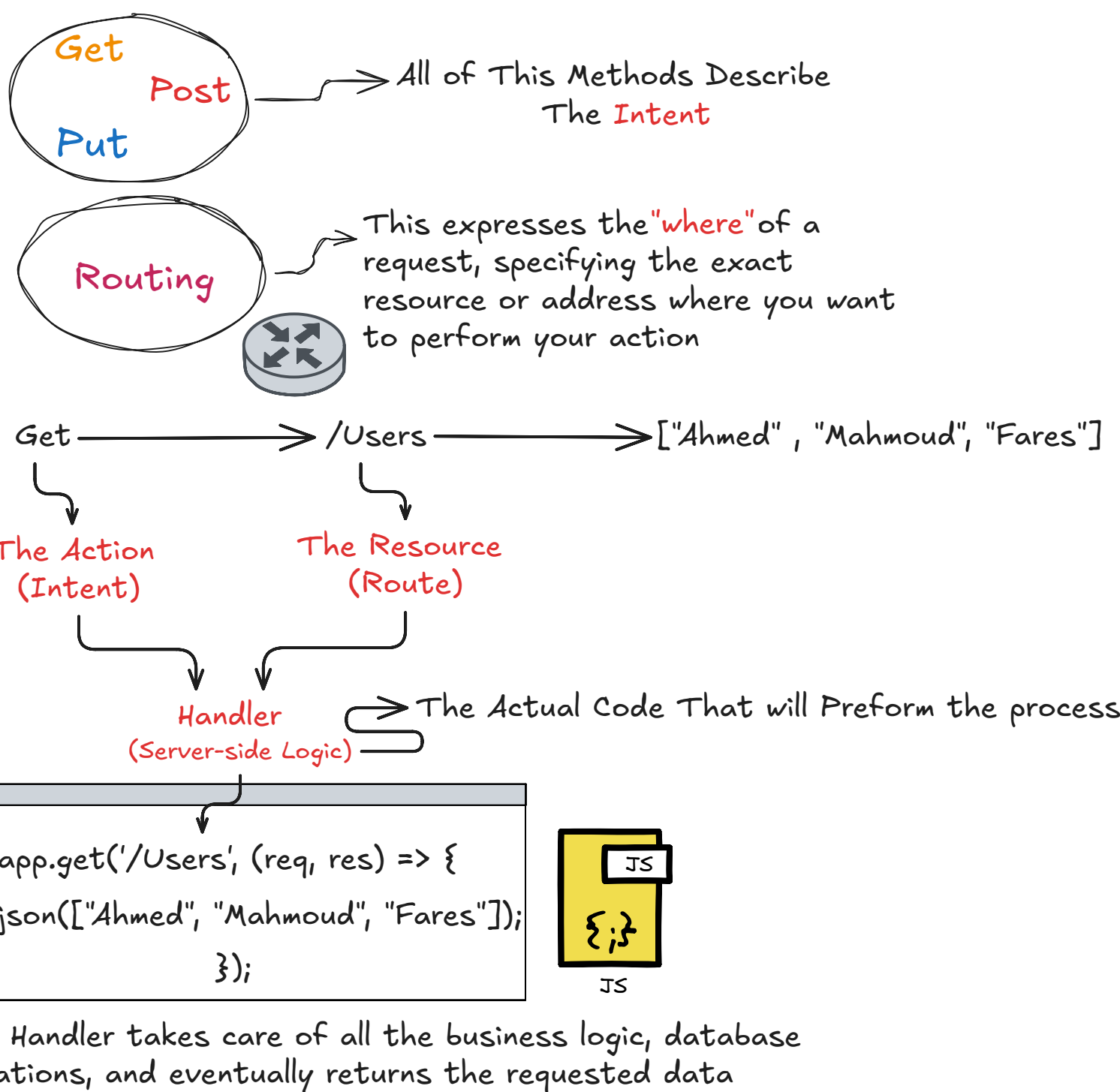


Routing in Backend



Route Types

Static Routes

These routes consist of a constant string that never changes and has no variable parameters

`/api/books`

`/api/users`

`/api/login`

`/api/books` is completely static; the address is consistent and it returns the same structural kind of response every time

Dynamic Routes `:id` → Path Parameter

These routes include variable slots, often denoted by a colon (e.g., `:id`). These are known as path parameters

For example,

in `/api/users/123`, the "123" is dynamically inserted into the `:id` slot, which the server extracts to fetch that exact user

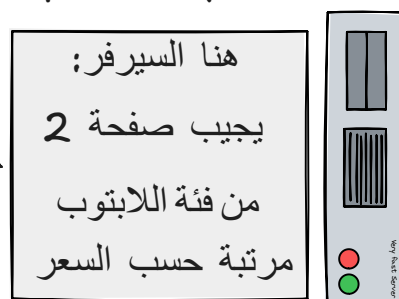
This creates a highly readable and semantic construct

Query Parameters `/api/search?key=value`

بيانات بنضيفها في نهاية ال URL بعد علامة ؟

Unlike path parameters, query parameters are appended to the very end of the route using a question mark and are formatted as key-value pairs

`/api/products?page=2&category=laptop&sort=price`



Nested Routes `/api/users/123/posts/456`

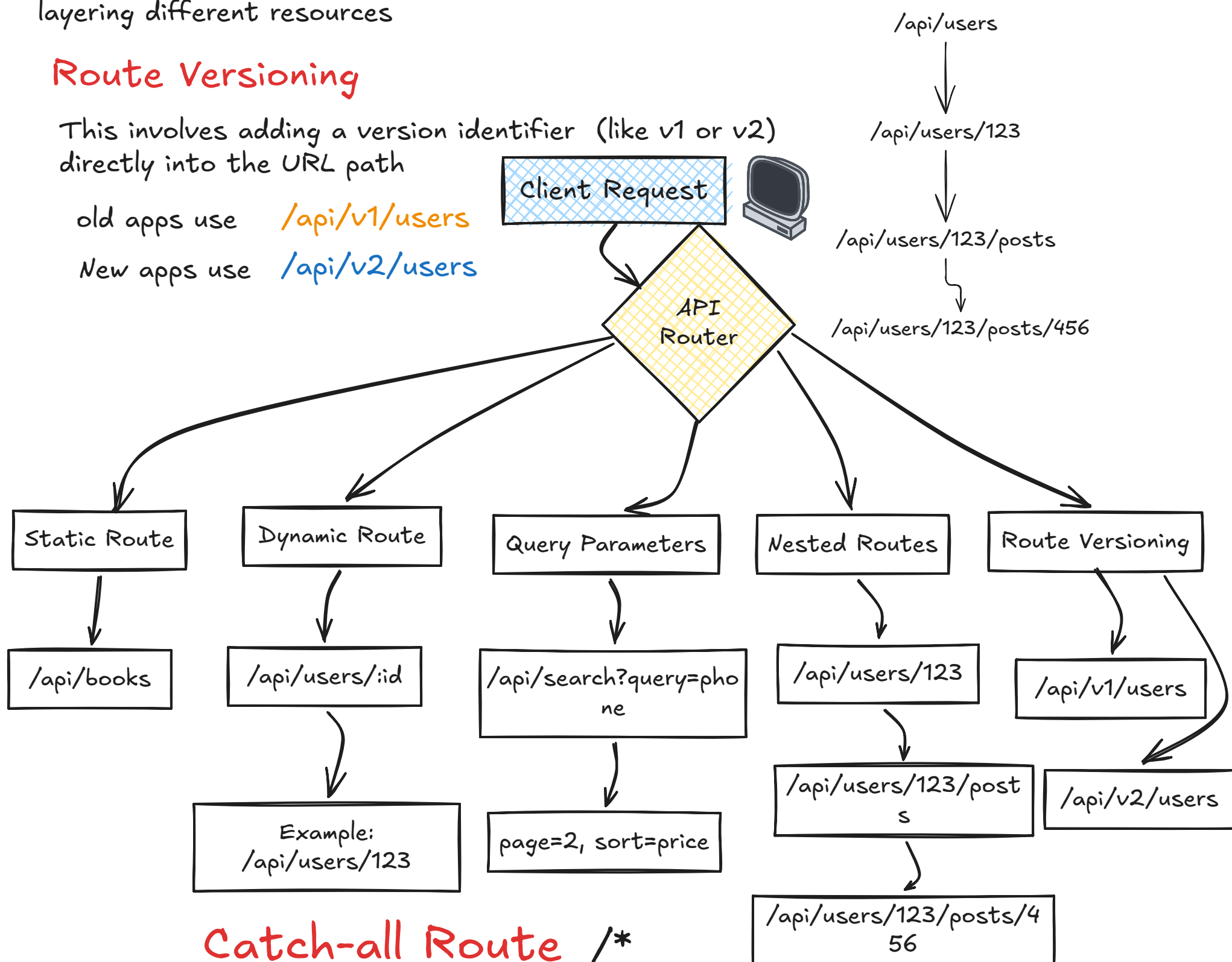
This is a standard REST API practice used to express semantic meaning by layering different resources

Route Versioning

This involves adding a version identifier (like `v1` or `v2`) directly into the URL path

old apps use `/api/v1/users`

New apps use `/api/v2/users`



Catch-all Route `/*`

